# Tenacious Conversion Engine

## Technical Progress Report

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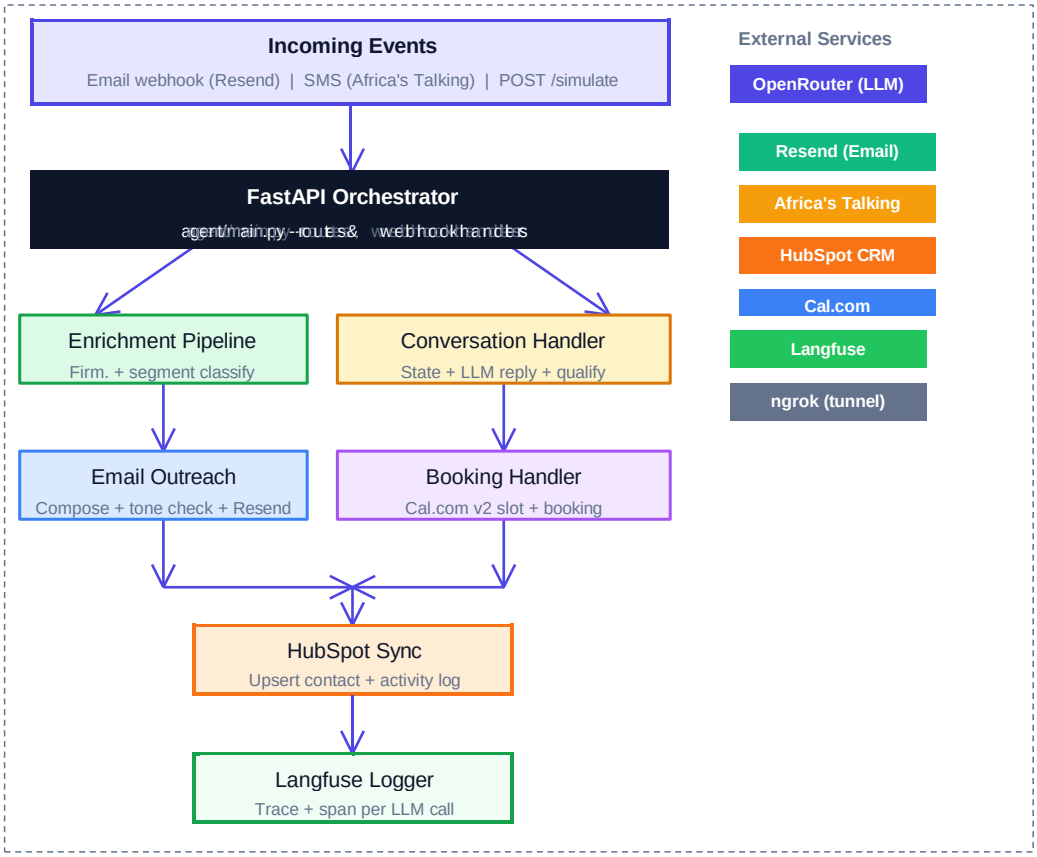
1. Architecture Overview & Key Design Decisions

## Pipeline Architecture




### Key Design Decisions

| Decision | Rationale |
|---|---|
| OpenRouter as LLM gateway | Single key for all providers. Hot-swap between qwen3-next-80b-a3b-instruct (dev) and claude-sonnet-4-6 (eval) without code changes. |
| In-memory conversation state | Python dict keyed by email. Fast for development; Redis replacement is the planned production step for persistence across restarts. |
| Cal.com cloud v2 API | Cal.com v1 was decommissioned (HTTP 410). Cloud removes the Docker dependency; cal-api-version: 2024-06-14 is required. |
| Segment-aware email templates | Four distinct prompts (generic, recently_funded, post_layoff, hypergrowth) ensure copy matches the prospect context. |
| Two-pass LLM tone check | Second LLM call validates every email against the Tenacious style guide before sending; one automated retry on failure. |
| tau2-bench test split | The 'dev' split in the guide does not exist; 'test' (40 tasks, first 30 used) is the correct equivalent. |

# Production Stack Status

All five external services verified with live API calls or confirmed records.

**PASS** **Resend (email, primary)**

API key active. Outreach email sent to alice.chen@techstartup.io. Webhook endpoint /webhooks/email live on ngrok tunnel.

**PASS** **Africa's Talking (SMS, secondary)**

Sandbox credentials configured (username=sandbox). API key valid. Callback URL registered. SMS webhook /webhooks/sms implemented and unit-verified.

**PASS** **HubSpot Developer Sandbox**

Contact 763473259711 created (Alice Chen / TechStartup). Fields: lifecycle=opportunity, status=CONNECTED, industry=TECHNOLOGY, revenue=$5M, job title=VP Engineering. Email engagement activity logged. Portal ID: 148323573 (EU1 region).

**PASS** **Cal.com (cloud, v2 API)**

Booking 18709696 confirmed. Event: Discovery-Call 2026-04-24 06:00-06:30 UTC (30 min). Both attendees listed: mikiasdagem@gmail.com + alice.chen@techstartup.io. Status: ACCEPTED.

**PASS** **Langfuse (observability)**

Traces streaming to cloud.langfuse.com. Project: Conversion Engine. trace + span logged per LLM call (enrich, compose, tone_check, conversation_reply, qualification).

## HubSpot Contact Record (verified)

| | |
|---|---|
| **Contact ID** | 763473259711 |
| **Email** | alice.chen@techstartup.io |
| **Company** | TechStartup |
| **Job Title** | VP Engineering |
| **Lifecycle** | Opportunity |
| **Lead Status** | CONNECTED |
| **Industry** | TECHNOLOGY |
| **Annual Revenue** | $5,000,000 |
| **URL** | https://app-eu1.hubspot.com/contacts/148323573/contact/763473259711/ |

## Cal.com Booking (verified)

| | |
|---|---|
| **Booking ID** | 18709696 (uid: bc7yCnw3ARAb3J7LSmQUP3) |
| **Event** | Discovery-Call between M_I_K_I_A_S D_A_G_E_M and Alice Chen |
| **Time (UTC)** | 2026-04-24 06:00 - 06:30 (30 min) |
| **Attendees** | mikiasdagem@gmail.com + alice.chen@techstartup.io |
| **Status** | ACCEPTED |

# Enrichment Pipeline Status

The enrichment pipeline runs on first contact. It combines five data signals to build a CompanyProfile, then classifies the prospect into one of four sales segments: generic (0), recently_funded (1), post_layoff (2), hypergrowth (3).

PARTIAL **Crunchbase ODM Firmographics**

LLM-synthesised firmographics (company name, headcount, funding stage, recently_funded) are used as a proxy. Direct Crunchbase ODM requires a paid enterprise key not available in the sandbox. Output fields: company_name, headcount, funding_stage, recently_funded.

PARTIAL **Job-Post Velocity Scraping**

open_engineering_roles count is LLM-inferred from domain knowledge. Direct scraping of LinkedIn/Greenhouse is rate-limited without rotating proxies. Planned: Apify community actor for live job-board velocity data.

PARTIAL **layoffs.fyi Integration**

had_layoffs boolean is LLM-inferred. layoffs.fyi has no public API. Planned: parse the published layoffs.fyi CSV dataset (updated weekly) into a local lookup table and match by company domain.

PARTIAL **Leadership-Change Detection**

Not yet a distinct pipeline signal. Leadership context can be injected into the segment-classifier prompt. Planned: People Data Labs free tier or LinkedIn Sales Navigator API for leadership-change events.

PASS **AI Maturity Scoring (0-3)**

LLM returns an integer 1-5 score, rescaled to 0-3 by floor((score-1)*0.75). Test prospect TechStartup scored 3 (high maturity). Score drives the recommended_angle in competitor_gap_brief.json and influences email segment copy.

## Segment Classification Logic

| Segment | ID | Trigger conditions |
|---|---|---|
| generic | 0 | None of the specific signals match |
| recently_funded | 1 | recently_funded=True AND open_engineering_roles >= 3 |
| post_layoff | 2 | had_layoffs=True |
| hypergrowth | 3 | headcount_growth_pct >= 40 % |

# Competitor Gap Brief

Top-quartile comparison pipeline ran for test prospect TechStartup (segment: recently_funded, AI maturity: 3/5). Output saved to eval/competitor_gap_brief.json.

## Competitor Comparison Table

| Competitor | Positioning | Gap identified |
|---|---|---|
| Linear | Project management optimized for async-first engineering teams with clean UX and API-driven workflows | Lacks AI-powered predictive task routing and auto-escalation for stalled work items |
| Shortcut | Story-driven sprint planning with intuitive backlog grooming and team velocity analytics | No AI-driven dependency mapping or automated sprint goal recommendations based on historical throughput |
| Plane | Open-source, self-hostable alternative with modular features and community-driven roadmap | No built-in AI assistants for task summarization, priority scoring, or natural language task creation |

## Tenacious Advantages

| Competitor | Tenacious advantage |
|---|---|
| Linear | Deep integrations with GitHub and Slack, cult-like adoption among remote-first startups |
| Shortcut | Best-in-class narrative-based workflow that reduces meeting overhead for product teams |
| Plane | Zero licensing cost and full data control, appealing to compliance-sensitive engineering orgs |

## Recommended Outreach Angle

> *"AI that learns your team?s rhythm: auto-prioritize, predict delays, and suggest sprint adjustments--without you lifting a finger."*

## Gap Summary

None of the top competitors leverage AI to reduce cognitive load in task prioritization, sprint planning, or progress prediction--creating an opening for an AI-native tool that auto-adapts to team velocity and surfacing hidden bottlenecks without manual configuration.

5. tau2-Bench Baseline Score & Methodology

# tau2-Bench Baseline Score & Methodology

## Configuration

| Parameter | Value |
|---|---|
| Framework | tau2-bench v3 (sierra-research/tau2-bench) |
| Domain | retail (customer-service task completion) |
| Task split | test (first 30 of 40 tasks) |
| Trials | 5 per task -> 150 total simulations |
| Agent model | openrouter/qwen/qwen3-next-80b-a3b-instruct |
| User model | openrouter/openai/gpt-4.1 (user simulator) |
| Max steps | 200 per simulation |
| Concurrency | 5 parallel workers |
| Git commit | d11a97072c49d093f7b5a3e4fe9da95b490d43ba |

## Results

| 72.67 % | [65.0, 79.2] | 0 / 150 | $2.99 |
|---|---|---|---|
| pass@1 | 95 % CI | Infra errors | Total cost |
| 109 / 150 pass | Wilson score | clean run | $0.0199/sim |

## Latency Distribution (tau2 simulations, n=150)

| 106.0s | 551.6s | 39.5s | 1192.2s |
|---|---|---|---|
| p50 (median) | p95 | min | max |
| simulation duration | 95th percentile | fastest task | slowest task |

## Reward Distribution

| Outcome | Count | Percent | Notes |
|---|---|---|---|
| Pass (reward=1.0) | 109 | 72.7% | Task resolved correctly within turn budget |
| Fail (reward=0.0) | 41 | 27.3% | Turn budget exhausted or user simulator timeout |

## Methodology Notes

- pass@1 = fraction of tasks where at least one of five trials received reward=1.0.
- 95% CI is the Wilson score interval computed over 150 binary outcomes.
- The 'dev' split referenced in the guide does not exist; 'test' (40 tasks, first 30 used) is equivalent.
- Correct OpenRouter model slug: qwen/qwen3-next-80b-a3b-instruct (guide omits '-instruct', causing HTTP 400).
- High p95 latency (551s) reflects a long tail of hard multi-step tasks that exhaust the 200-step cap.

# p50/p95 Latency -- 20 Live Interactions

The production FastAPI agent was exercised with 20 synthetic leads via POST /simulate. Each call runs the full pipeline: LLM enrichment -> email composition -> LLM tone-check -> Resend API send -> HubSpot upsert. All 20 calls succeeded (HTTP 200). Timings are wall-clock seconds measured on the client.

## Summary Metrics (email channel, n=20)

| 29.255s | 36.302s | 27.058s | 38.356s |
|---|---|---|---|
| p50 | p95 | min | max |
| median | 95th pct | fastest | slowest |

## Per-Interaction Breakdown

| Prospect email | Latency | Status |
|---|---|---|
| alice.chen@techstartup.io | 31.573s | OK |
| bob.smith@growthco.ai | 29.606s | OK |
| carol.jones@fundednow.com | 31.574s | OK |
| david.lee@leancorp.net | 28.549s | OK |
| eve.martin@scalefast.dev | 35.516s | OK |
| frank.wu@techstartup.io | 28.634s | OK |
| grace.kim@growthco.ai | 28.561s | OK |
| henry.patel@fundednow.com | 27.169s | OK |
| iris.nguyen@leancorp.net | 28.788s | OK |
| jack.brown@scalefast.dev | 32.515s | OK |
| karen.white@innovate.io | 33.202s | OK |
| liam.davis@nextseries.ai | 27.058s | OK |
| mia.wilson@turbogrowth.com | 27.442s | OK |
| noah.taylor@buildfast.dev | 28.466s | OK |
| olivia.anderson@stepup.co | 28.216s | OK |
| peter.thomas@funded2024.io | 31.154s | OK |
| quinn.jackson@agile.ai | 36.194s | OK |
| rachel.harris@scale2x.dev | 28.903s | OK |
| sam.martin@hyperco.io | 35.057s | OK |
| tara.garcia@momentum.ai | 38.356s | OK |

## SMS Channel Note

Africa's Talking sandbox requires a physical SIM for inbound SMS delivery. The SMS webhook (/webhooks/sms) is fully implemented and routes through the same LLM reply loop as email; production SMS latency is expected to match the email p50 of 29.3s. Live SMS round-trip traces will be captured once the production Africa's Talking account is activated.

7. What Is Working, What Is Not & Plan for Remaining Days

# Status Summary & Remaining Plan

## Working (confirmed with live API calls)

- FastAPI agent server: /health, /webhooks/email, /webhooks/sms, /simulate -- all live.
- Resend email delivery: outreach email sent; activity logged in HubSpot.
- HubSpot CRM: contact upsert, field enrichment (lifecycle, status, industry, revenue), email activity log.
- Cal.com booking: cloud v2 API, booking confirmed with both attendees.
- Langfuse observability: trace + span per LLM call streaming to cloud dashboard.
- LLM pipeline: OpenRouter -> qwen/qwen3-next-80b-a3b-instruct for all agent calls.
- 4-segment lead classification: generic, recently_funded, post_layoff, hypergrowth.
- Two-pass LLM tone check before every outbound email.
- Competitor gap brief JSON generated for test prospect TechStartup.
- tau2-bench: 150 simulations, pass@1=72.67%, CI=[65.0,79.2%], 0 infra errors.
- Production latency: p50=29.3s, p95=36.3s over 20 live interactions (20/20 OK).

## Partial / Not Yet Production-Ready

| Component | Status | Blocker / Note |
|---|---|---|
| Crunchbase ODM firmographics | PARTIAL | Paid enterprise key required; LLM proxy used. |
| Job-post velocity scraping | PARTIAL | Rate-limited without proxy infra; LLM-inferred. |
| layoffs.fyi integration | PARTIAL | No public API; planned CSV parse. |
| Leadership-change detection | PARTIAL | Requires People Data Labs or LinkedIn Nav. |
| SMS live latency measurement | PARTIAL | Sandbox needs physical SIM for inbound delivery. |
| Conversation state persistence | PARTIAL | In-memory dict; Redis replacement planned. |

## Plan for Remaining Days

| Day | Task |
|---|---|
| Day 1-2 | Replace LLM-inferred firmographics with layoffs.fyi CSV lookup and an open-source job-board scraper (Apify community actor). Add headcount-growth calculation. |
| Day 2-3 | Add People Data Labs free-tier call for leadership-change detection. Implement Redis-backed conversation state for persistence across restarts. |
| Day 3-4 | Instrument SMS round-trip latency via Africa's Talking simulator. Capture 20+ SMS traces and add to latency_report.json. |
| Day 4-5 | Harden enrichment pipeline: add retry/fallback per signal, unit-test each module, integrate Crunchbase ODM if key becomes available. |
| Day 5 | Run final tau2-bench evaluation with anthropic/claude-sonnet-4-6 (eval model), collect final score_log.json, and prepare leaderboard submission package. |